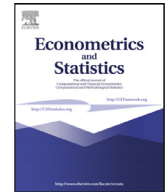




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Empirical best predictors under multivariate Fay-Herriot models and their numerical approximation

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ABSTRACT

Small area estimation of multivariable non-linear domain indicators using aggregated data is addressed. By assuming that the target vector follows a multivariate Fay-Herriot model, empirical best predictors of domain parameters that are arbitrary Lebesgue-measurable functions of multiple target variables are derived. In this context, Monte Carlo and Gauss-Hermite quadrature methods for integral approximation are discussed. A parametric bootstrap algorithm for mean squared error estimation is presented. Simulation experiments are conducted to study the behaviour of the introduced methodology. Moreover, an illustrative application to real data from the Spanish labour force survey is provided. In this example, province-level unemployment rates, crossed by age classes and sex, are estimated.

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1. Introduction

National statistical plans often need to estimate domain indicators that are non-linear functions of two or more target variables. Examples are regional expenditure ratios, income shares, and unemployment rates. Considering ratio estimation, the simplest approach is to estimate the numerator and denominator independently, and then plug the results into the ratio formula (plug-in estimators). However, this approach has several disadvantages. First, it does not consider the joint distribution of the target variables, which may lead to inefficiency when the target variables are correlated. Second, a corresponding plug-in estimator is biased due to Jensen's inequality. Moreover, asymptotic unbiasedness cannot be assumed since domain-level indicator estimates are typically affected by small samples due to limited survey resources.

Small area estimation (SAE) addresses the estimation of domain indicators in the presence of small samples. In these settings, direct estimators suffer from high uncertainty due to large sampling variances. SAE solves this problem by combining information from multiple domains within statistical models considering suitable auxiliary variables. Domain indicator estimates are then obtained via model prediction. We refer to [Rao and Molina \(2015\)](#) as well as [Morales et al. \(2021\)](#) for a comprehensive overview of SAE. Statistical models in SAE are commonly fitted on either unit- or area-level data. For a unit-level model, the estimation of non-linear indicators requires auxiliary variable records to be available for the entire population, e.g. obtained from a census. Naturally, this severely limits the applicability of the unit-level approach in practice.

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In contrast, area-level models only require aggregated data records on the domain-level which are much easier to obtain. Under this approach, domain indicators are explicit functions of the model parameters. Accordingly, a suitable model specification is essential for constructing estimators with good statistical properties. For instance, under linear mixed models (LMMs), we can apply empirical best linear unbiased predictors (EBLUPs) to estimate linear indicators. However, as initially mentioned, domain indicators are often non-linear, which prevents us from using the EBLUP. Instead, empirical best predictors (EBPs) should be applied, as they allow for arbitrary indicators. We therefore propose to use multivariate LMMs to construct EBPs of domain indicators that are non-linear functions of multiple target variables.

In this field, the pioneering work of Fay (1987), Datta et al. (1991), and Datta et al. (1999) inspired many research articles on either multivariate modelling or EBPs of non-linear indicators. In what follows, we address some selected contributions without being exhaustive. Multivariate Fay-Herriot (MFH) models were employed by Morales et al. (2015), Porter et al. (2015), Benavent and Morales (2016, 2021), Arima et al. (2017), and Burgard et al. (2021) to estimate poverty indicators. Likewise, they were used by Marchetti and Secondi (2017) as well as Ubaidillah et al. (2019) to estimate household consumption expenditures. Erciulescu and Opsomer (2019) predicted employee compensation components by using hierarchical Bayes bivariate Fay-Herriot (BFH) models. Burgard et al. (2022b) extended the BFH model to partially-missing values in the target variables. MFH models for compositional data were applied by Esteban et al. (2020) and Krause et al. (2022). With respect to EBPs, Jiang and Lahiri (2001) and Jiang (2003) introduce EBPs of non-linear parameters under generalized linear mixed models. Following their approach, Boubeta et al. (2016, 2017) and Faltys et al. (2022) derived EBPs of small area poverty proportions based on area-level Poisson and Gamma mixed models, respectively. Molina et al. (2007), López-Vizcaíno et al. (2013, 2015) assumed area-level multinomial mixed models for estimating labour indicators. Chandra et al. (2017, 2018) as well as Chandra and Salvati (2018) introduced predictors of spatially correlated area-level count data. Under a benchmarking approach, Erciulescu et al. (2018) introduced a procedure for estimating ratio parameters.

These contributions show that the area-level SAE approach has recently received considerable attention, mainly for the estimation of univariate linear indicators using EBLUPs. However, we found that the estimation of non-linear indicators that depend on multiple target variables has not been addressed with EBP using multivariate LMMs so far. The calculation of these EBPs requires the approximation of multiple integrals that can have a high computational cost, because the approximation methods do not have high convergence rates in the multivariate case. In this sense, it is necessary to investigate the most efficient way, with low estimation error and moderate computational cost, to calculate an EBP. This paper try to close a theoretical gap by presenting the appropriate mathematical developments and providing practical computational information.

Based on the MFH model, we introduce EBPs of non-linear domain indicators that are functions of multiple target variables. As the EBPs contain integrals that cannot be solved analytically, we study different numerical approximation techniques, such as the Monte Carlo and Gauss-Hermite quadrature methods. Parametric bootstrap estimators for MSE estimation are presented. A Fisher-scoring algorithm is used to obtain residual maximum likelihood (REML) estimators of the model parameters. Simulation experiments are conducted to evaluate the effectiveness of the proposed methodology. We further present an illustrative application to the Spanish labour force data, where the target is the estimation of small area unemployment rates of Spanish provinces, crossed by age classes and sex.

The rest of the paper is organized as follows. Section 2 introduces the MFH model. Section 3 derives the EBPs of indicators that depend on multiple target variables. It also presents some methods for numerical integration. Section 4 describes parametric bootstrap procedures to estimate MSEs of the EBPs. Section 5 carries out simulation experiments to investigate the behaviour of the EBPs and the MSE estimators. Section 6 gives the application to real data. Section 7 presents some conclusions. The manuscript is supported by a supplementary material with seven appendices. Appendix A contains mathematical proofs. Appendix B describes the REML estimation method for MFH models. Appendix C deals with the prediction of unemployment rates. Appendix D reviews some integration techniques for approximating the EBPs. Appendix E contains complementary simulation results and the details of the simulation set-ups. Appendix F presents further insights on the application to real data. Appendix G provides a particularization of the EBPs for the case that the target indicator is a function of one random variable.

2. The multivariate Fay-Herriot model

This section introduces the multivariate Fay-Herriot (MFH) model as basis for the EBP developments. Note that the proofs of the developments are located in Appendix A. Let U be a finite population partitioned into D domains $U_1, \dots, U_D$. Let $\mu_d = (\mu_{d1}, \dots, \mu_{dm})'$ be a vector of characteristics of interest in the domain d and let $y_d = (y_{d1}, \dots, y_{dm})'$ be a vector of direct estimators of μ_d calculated by using the data of the target survey sample. The MFH model is defined in two stages. The first stage indicates that direct estimators y_d , $d = 1, \dots, D$, are unbiased and follow the *sampling model*

$$y_d = \mu_d + e_d, \quad d = 1, \dots, D, \quad (1)$$

where the vectors $e_d = (e_{d1}, \dots, e_{dm})' \sim N(0, V_{ed})$ are independent and the $m \times m$ covariance matrices V_{ed} are known. In most cases, V_{ed} is taken to be the design-based covariance matrix of the direct estimator y_d , $d = 1, \dots, D$. The covariance

matrices V_{ed} are

$$V_{ed} = \begin{pmatrix} \sigma_{ed1}^2 & \sigma_{ed12} & \cdots & \sigma_{ed1m} \\ \sigma_{ed12} & \sigma_{ed2}^2 & \cdots & \sigma_{ed2m} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{ed1m} & \sigma_{ed2m} & \cdots & \sigma_{edm}^2 \end{pmatrix}, \quad d = 1, \dots, D.$$

In the second stage, the true area characteristic μ_{dk} is assumed to be linearly related to p_k explanatory variables, $k = 1, \dots, m$, $d = 1, \dots, D$. Let $x'_{dk} = (x_{dk1}, \dots, x_{dkp_k})$ be a row vector containing the true aggregated (population) values of p_k explanatory variables for μ_{dk} and let $X_d = \text{diag}(x'_{d1}, \dots, x'_{dm})$ be a full rank $m \times p$ block-diagonal matrix with $p = p_1 + \dots + p_m$. Let $\beta_k = (\beta_{k1}, \dots, \beta_{kp_k})'$ be a column vector of size p_k containing the regression parameters for μ_{dk} , $k = 1, \dots, m$, and let $\beta = (\beta'_1, \dots, \beta'_m)'_{p \times 1}$. The linking model is

$$\mu_d = X_d \beta + u_d, \quad u_d = (u_{d1}, \dots, u_{dm})' \sim N_m(0, V_{ud}), \quad d = 1, \dots, D, \quad (2)$$

where the vectors u_d 's and e_d 's are all mutually independent. The $m \times m$ covariance matrix V_{ud} depends on $q = m(m+1)/2$ unknown parameters denoted by

$$\theta = (\theta_1, \dots, \theta_q)' = (\sigma_{u1}^2, \sigma_{u2}^2, \dots, \sigma_{um}^2, \rho_{12}, \rho_{13}, \dots, \rho_{23}, \rho_{24}, \dots, \rho_{m-1,m})'$$

and is given by

$$V_{ud} = \begin{pmatrix} \sigma_{u1}^2 & \rho_{12}\sigma_{u1}\sigma_{u2} & \cdots & \rho_{1m}\sigma_{u1}\sigma_{um} \\ \rho_{12}\sigma_{u1}\sigma_{u2} & \sigma_{u2}^2 & \cdots & \rho_{2m}\sigma_{u2}\sigma_{um} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1m}\sigma_{u1}\sigma_{um} & \rho_{2m}\sigma_{u2}\sigma_{um} & \cdots & \sigma_{um}^2 \end{pmatrix}.$$

The MFH model can be expressed as a single model in the form

$$y_d = X_d \beta + u_d + e_d, \quad d = 1, \dots, D, \quad (3)$$

or in the linear mixed model matrix form

$$y = X\beta + Zu + e,$$

where

$$y = \text{col}_{1 \leq d \leq D}(y_d), \quad u = \text{col}_{1 \leq d \leq D}(u_d), \quad e = \text{col}_{1 \leq d \leq D}(e_d), \quad X = \text{col}_{1 \leq d \leq D}(X_d), \quad Z = I_{mD}.$$

Under model (3), it holds that $y \sim N(X\beta, V)$ with

$$E[y] = X\beta \quad \text{and} \quad V = \text{var}(y) = Z'V_uZ + V_e = V_u + V_e = \text{diag}_{1 \leq d \leq D}(V_d),$$

where

$$V_u = V_u(\theta) = \text{diag}_{1 \leq d \leq D}(V_{ud}), \quad V_e = \text{diag}_{1 \leq d \leq D}(V_{ed}), \quad V_d = V_{ud} + V_{ed}, \quad d = 1, \dots, D.$$

Based on this model structure, we can now derive the best predictors (BPs) under the model. Thereby, the following lemma gives a useful relationship between the variance matrices V_d , V_{ud} , and V_{ed} .

Lemma 2.1 *It holds that*

$$V_{ud}V_d^{-1} = (V_{ed}^{-1} + V_{ud}^{-1})^{-1}V_{ed}^{-1}.$$

The proof can be found in Appendix A (supplemental material). The next proposition gives the posterior distribution of u_d , given y_d . Based on this result, we can derive the best predictors (BPs) of u_d and μ_d , $d = 1, \dots, D$.

Proposition 2.1 *The posterior distribution of u_d , given y_d , is multivariate normal with mean vector and variance matrix*

$$\hat{u}_d^{bp} = E[u_d | y_d; \beta, \theta] = \Phi_d V_{ed}^{-1} (y_d - X_d \beta) = V_{ud} V_d^{-1} (y_d - X_d \beta),$$

$$\Phi_d = \text{var}(u_d | y_d; \beta, \theta) = (V_{ed}^{-1} + V_{ud}^{-1})^{-1} = V_{ud} V_d^{-1} V_{ed}.$$

The proof can be found in Appendix A (supplemental material). Recall that the BP of an unknown quantity in domain U_d under the MFH model is its conditional expectation given y_d under the assumption of known model parameters. Thus, for given β and θ , the BPs of u and μ are

$$\hat{u}^{bp} = V_u V^{-1} (y - X\beta), \quad \hat{\mu}^{bp} = X\beta + \hat{u}^{bp}. \quad (4)$$

If θ is known and β is unknown, then best linear unbiased estimator (BLUE) of β and the best linear unbiased predictor (BLUP) of u and $\mu = X\beta + Zu$ are

$$\hat{\beta}^{blue} = (X'V^{-1}X)^{-1}X'V^{-1}y, \quad \hat{u}^{blup} = V_uV^{-1}(y - X\hat{\beta}^{blue}), \quad \hat{\mu}^{blup} = X\hat{\beta}^{blue} + Z\hat{u}^{blup}. \quad (5)$$

By substituting θ in the formulas (5) by an estimator $\hat{\theta}$, we obtain the empirical BLUE (EBLUE) of β and the empirical BLUP (EBLUP) of u and $\mu = X\beta + Zu$, i.e.

$$\hat{\beta} = (X'\hat{V}^{-1}X)^{-1}X'\hat{V}^{-1}y, \quad \hat{u}^{eblup} = \hat{V}_u\hat{V}^{-1}(y - X\hat{\beta}), \quad \hat{\mu}^{eblup} = X\hat{\beta} + Z\hat{u}^{eblup}, \quad (6)$$

where $\hat{V}_u = V_u(\hat{\theta})$ and $\hat{V} = \hat{V}_u + V_e$ are obtained by plugging $\hat{\theta}$ in the place of θ . This paper substitutes θ by its REML estimator $\hat{\theta}$ and denotes the complete vector of model parameters as $\psi = (\beta', \theta')'$. Appendix B develops the REML estimation method for MFH models, gives the asymptotic distributions of the REML estimators and presents confidence intervals and test statistics. Moreover, note that suitable criteria for model selection that based on the goodness of fit in model parameter estimation can be found in [Jiang et al. \(2008\)](#), [Shang et al. \(2008\)](#), as well as [Rao \(2014\)](#).

3. Prediction of domain parameters

This section presents the developments on domain parameter prediction and discusses numerical integration methods. We refer to [Press et al. \(2007\)](#) for an overview of numerical integration techniques, [Stroud and Secrest \(1966\)](#) for Gaussian quadrature formulas, and [Lange \(2010\)](#) as well as [Fishman \(1996\)](#) for Monte Carlo integration techniques. Corresponding methodologies have recently been applied in the SAE context, e.g., by [Sugasawa et al. \(2017\)](#) and [Sun et al. \(2022\)](#).

The domain parameters of interest take the form

$$G_d = G_d(u_d, \beta) = g(\mu_{d1}, \dots, \mu_{dm}), \quad d = 1, \dots, D,$$

where $g: \mathbb{R}^m \mapsto \mathbb{R}$ is a Lebesgue-measurable function. Examples of parameters G_d , that depend on one, two or more μ_{dk} 's, are totals of unemployed, unemployment rates, and ratios of food household expenditures. The BP and the plug-in predictor (IN) of G_d are given by

$$\hat{G}_d^{bp} = E_{\psi}[G_d|y_d] = \int_{\mathbb{R}^m} G_d(u_d, \beta) f_{\psi}(u_d|y_d) du_d, \quad (7)$$

$$\hat{G}_d^{in} = G_d(\hat{u}_d^{bp}, \beta). \quad (8)$$

Observe that the BP is unbiased by construction, whereas the plug-in is biased due to Jensen's inequality. The EBP of G_d is obtained by substituting $\psi = (\beta', \theta')'$ by an estimator in the formula (7). This paper considers the REML estimator $\hat{\psi}$, so the EBP of G_d is

$$\hat{G}_d^{ebp} = E_{\hat{\psi}}[G_d|y_d], \quad d = 1, \dots, D. \quad (9)$$

The empirical plug-in predictor (EIN) of G_d is obtained by substituting μ_d by the EBLUP $\hat{\mu}_d^{eblup}$, i.e.

$$\hat{G}_d^{ein} = g\left(\hat{\mu}_{d1}^{eblup}, \dots, \hat{\mu}_{dm}^{eblup}\right), \quad d = 1, \dots, D. \quad (10)$$

As the simulations in [Section 5](#) consider on domain unemployment rates as an example of a non-linear indicator, note that Appendix C shows how to calculate the EBPs and the plug-in predictors of domain unemployment rates by Monte Carlo integration.

In general, the m -dimensional integral (7) is not analytically calculable and can therefore only be approximated numerically. We consider different forms of the integral of the BP of G_d , namely integral forms I1, I1., I2, and I3 described below. To approximate the multi-dimensional integrals, a range of different algorithms can be used such as quadrature methods, Monte Carlo (MC) integration, and Quasi-Monte Carlo (QMC) integration. Appendix D gives a short overview of different methods and an example of Gauss-Hermite quadrature (GHQ) for a multivariate normal distribution.

This paper discusses the prediction of non-linear functions of area-level quantities by MFH models. We expect the m -dimensional integral of (7) to be of lower order, e.g. $m \leq 5$, in practical applications of MFH models and therefore an application of GHQ with product rule to be feasible. We present some approximations to the m -dimensional integral (7) by GHQ and MC integration (including antithetic variates). With the given literature in Appendix D, it is straightforward to extend the following integral approximations to sparse-grid formulas, QMC integration, or with other variance reduction techniques in MC integration, to account for the features of the specific non-linear statistic of interest.

In what follows, we present different integral forms and approximations of (7). For ease of exposition, we use the same notation for the BPs in their integral form and for the approximated BPs, i.e. the approximations of the integral. This is to say, they are both denoted by $\hat{G}_d^{bp}$. Similarly, EBPs and approximated EBPs are denoted in the same way by $\hat{G}_d^{ebp}$. We recall that we can only calculate the approximated EBPs in practice. We present the integral forms and approximations to the BP of G_d . The approximations to the EBP of G_d are obtained by substituting ψ by an estimator.

Integral form I1. As the conditional distribution is

$$f_{\psi}(u_d|y_d) = \frac{f_{\psi}(u_d, y_d)}{f_{\psi}(y_d)} = \frac{f_{\psi}(y_d|u_d)f_{\theta}(u_d)}{f_{\psi}(y_d)}, \quad d = 1, \dots, D,$$

the BP of G_d is

$$\begin{aligned} \hat{G}_d^{bp} &= \int_{\mathbb{R}^m} G_d(u_d, \beta) \frac{f_{\psi}(y_d|u_d)f_{\theta}(u_d)}{f_{\psi}(y_d)} du_d = \frac{\int_{\mathbb{R}^m} G_d(u_d, \beta) f_{\psi}(y_d|u_d)f_{\theta}(u_d) du_d}{f_{\psi}(y_d)} \\ &= \frac{\int_{\mathbb{R}^m} G_d(u_d, \beta) f_{\psi}(y_d|u_d)f_{\theta}(u_d) du_d}{\int_{\mathbb{R}^m} f_{\psi}(y_d|u_d)f_{\theta}(u_d) du_d} = \frac{A_d(y_d, \beta, \theta)}{B_d(y_d, \beta, \theta)}. \end{aligned} \quad (11)$$

where

$$\begin{aligned} A_d(y_d, \beta, \theta) &= \int_{\mathbb{R}^m} G_d(u_d, \beta) \exp \left\{ -\frac{1}{2} r_d' V_{ed}^{-1} r_d \right\} f_{\theta}(u_d) du_d, \\ B_d(y_d, \beta, \theta) &= \int_{\mathbb{R}^m} \exp \left\{ -\frac{1}{2} r_d' V_{ed}^{-1} r_d \right\} f_{\theta}(u_d) du_d, \quad r_d = r_d(u_d) = y_d - X_d \beta - u_d. \end{aligned}$$

Algorithm 1_{mc} gives a Monte Carlo approximation (with antithetic variates) to the integral form (11) of $\hat{G}_d^{bp}$. For $d = 1, \dots, D$, the steps are

1. For $t = 1, \dots, T$, draw $u_d^{(t)} \stackrel{iid}{\sim} N_m(0, V_{ud}(\theta))$ and set $u_d^{(T+t)} = -u_d^{(t)}$.
2. Approximate $\hat{G}_d^{bp} \approx \frac{\hat{A}_d(y_d, \beta, \theta)}{\hat{B}_d(y_d, \beta, \theta)}$, where

$$\begin{aligned} \hat{A}_d(y_d, \beta, \theta) &= \frac{1}{2T} \sum_{t=1}^{2T} G_d(u_d^{(t)}, \beta) \exp \left\{ -\frac{1}{2} r_d^{(t)'} V_{ed}^{-1} r_d^{(t)} \right\}, \\ \hat{B}_d(y_d, \beta, \theta) &= \frac{1}{2T} \sum_{t=1}^{2T} \exp \left\{ -\frac{1}{2} r_d^{(t)'} V_{ed}^{-1} r_d^{(t)} \right\}, \quad r_d^{(t)} = y_d - X_d \beta - u_d^{(t)}. \end{aligned}$$

Algorithm 1_{ghq} gives a GHQ approximation to the integral form (11) of $\hat{G}_d^{bp}$. For $d = 1, \dots, D$, the steps are

1. Decompose $V_{ud}(\theta) = L_d L_d'$ and set $z_d = 2^{-0.5} L_d^{-1} u_d$, $u_d = \sqrt{2} L_d z_d$ and $r_d^* = y_d - X_d \beta - \sqrt{2} L_d z_d$. For $K_d \in \{A_d, B_d\}$, we have

$$\begin{aligned} K_d(y_d, \beta, \theta) &= \pi^{-m/2} \int_{\mathbb{R}^m} h_{Kd}(z_d) \exp \{ -z_d' z_d \} dz_d, \\ h_{Bd}(z_d) &= \exp \left\{ -\frac{1}{2} r_d^{*'} V_{ed}^{-1} r_d^* \right\}, \quad h_{Ad}(z_d) = G_d(\sqrt{2} L_d z_d, \beta) h_{Bd}(z_d). \end{aligned}$$

2. Approximate $\hat{G}_d^{bp} \approx \frac{\hat{A}_d(y_d, \beta, \theta)}{\hat{B}_d(y_d, \beta, \theta)}$, where

$$\hat{K}_d(y_d, \beta, \theta) = \sum_{t_1=1}^T \dots \sum_{t_m=1}^T \pi^{-m/2} h_{\hat{K}d}(\tilde{z}_{t_1, \dots, t_m}) \prod_{k=1}^m w_{t_k}, \quad \hat{K}_d \in \{\hat{A}_d, \hat{B}_d\},$$

with $\tilde{z}_{t_1, \dots, t_m} = \sqrt{2} L_d (z_{t_1}, \dots, z_{t_m})'$, where w_{t_k} and z_{t_k} are the standard GHQ weights and nodes for degree T , $t_k = 1, \dots, T$, $k = 1, \dots, m$.

Integral form I1 *. The denominator in formula (11) is $f_{\psi}(y_d)$ and $y_d \sim N_m(X_d \beta, V_d)$. Therefore, we have the alternative integral form

$$\hat{G}_d^{bp} = \frac{|V_d(\theta)|^{1/2} A_d(y_d, \beta, \theta)}{|V_{ed}(\theta)|^{1/2} B_d^*(y_d, \beta, \theta)}, \quad (12)$$

where

$$B_d^*(y_d, \beta, \theta) = (2\pi)^{m/2} |V_d|^{1/2} f_{\psi}(y_d) = \exp \left\{ -\frac{1}{2} r_d' V_d^{-1} r_d \right\}.$$

Algorithms 1_{mc}^{*} and 1_{ghq}^{*} give MC and GHQ approximations to the BP of G_d by running the same steps as Algorithm 1_{mc} and 1_{ghq}, but applying the formula (12) respectively.

Integral form I2. From formula (7), the BP of G_d is given by

$$\hat{G}_d^{bp} = E_{\psi}[G_d|y_d] = \int_{\mathbb{R}^m} G_d(u_d, \beta) f_{\psi}(u_d|y_d) du_d.$$

Algorithm 2_{mc} gives a Monte Carlo approximation to the integral form (7) of $\hat{G}_d^{bp}$. For $d = 1, \dots, D$, the steps are:

1. For $t = 1, \dots, T$, draw $u_d^{(t)} \stackrel{iid}{\sim} N_m(\mu_{u_d|y_d}(\theta), V_{u_d|y_d}(\theta))$, where $\mu_{u_d|y_d}(\beta, \theta) = V_{ud}(\theta)V_d^{-1}(\theta)(y_d - X_d\beta)$, $V_{u_d|y_d}(\theta) = V_{ud}(\theta)V_d^{-1}(\theta)V_{ed}$, and set $u_d^{(T+t)} = 2\mu_{u_d|y_d}(\beta, \theta) - u_d^{(t)}$.
2. $\hat{G}_d^{bp} \approx \frac{1}{2T} \sum_{t=1}^{2T} G_d^{(t)}$ where $G_d^{(t)} = G_d(u_d^{(t)}, \beta)$.

Algorithm 2_{ghq} gives a GHQ approximation to the integral form (7) of $\hat{G}_d^{bp}$. For $d = 1, \dots, D$, the steps are

1. For $f_\psi(u_d|y_d)$ we have $\mu_{u_d|y_d}(\beta, \theta) = V_{ud}(\theta)V_d^{-1}(\theta)(y_d - X_d\beta)$ and $V_{u_d|y_d}(\theta) = V_{ud}(\theta)V_d^{-1}(\theta)V_{ed}$.
2. Take decomposition $V_{u_d|y_d}(\theta) = L_d L_d'$ and $z_d = 2^{-0.5} L_d^{-1}(u_d - \mu_{u_d|y_d})$, $u_d = \sqrt{2} L_d z_d + \mu_{u_d|y_d}$, such that

$$\hat{G}_d^{bp} = \pi^{-m/2} \int_{\mathbb{R}^m} G_d(\sqrt{2} L_d z_d + \mu_{u_d|y_d}, \beta) \exp \left\{ -z_d' z_d \right\} dz_d.$$

3. Approximate

$$\hat{G}_d^{bp} \approx \sum_{t_1=1}^T \dots \sum_{t_m=1}^T \pi^{-m/2} G_d(\tilde{z}_{t_1, \dots, t_m}) \prod_{k=1}^m w_{t_k},$$

with $\tilde{z}_{t_1, \dots, t_m} = \sqrt{2} L_d (z_{t_1}, \dots, z_{t_m})' + \mu_{u_d|y_d}$, where w_{t_k} and z_{t_k} are the standard GHQ weights and nodes for degree T , $k = 1, \dots, m$, $t = 1, \dots, T$.

Integral form I3. Recall the first integral in (11), i.e.

$$\hat{G}_d^{bp} = \int_{\mathbb{R}^m} G_d(u_d, \beta) \frac{f_\psi(y_d|u_d)}{f_\psi(y_d)} f_\theta(u_d) du_d.$$

Algorithm 3_{mc} gives a sampling importance resampling (SIR) Monte Carlo approximation to I3. See e.g. [Smith and Gelfand \(1992\)](#) for a description of the SIR algorithm in the Bayesian context. For $d = 1, \dots, D$, the steps of Algorithm 3_{mc} are

1. For $t = 1, \dots, T$, $s = 1, \dots, S$, draw $u_d^{(s,t)} \stackrel{iid}{\sim} N_m(0, V_{ud}(\theta))$ and set $u_d^{(S+s,t)} = -u_d^{(s,t)}$.
2. For $t = 1, \dots, T$, set $u_d^{(t)} = u_d^{(s,t)}$ with probability

$$p_d^{(s,t)} \propto f(y_d|u_d^{(s,t)}) \propto \exp \left\{ -\frac{1}{2} (y_d - X_d\beta - u_d^{(s,t)})' V_{ed}^{-1} (y_d - X_d\beta - u_d^{(s,t)}) \right\}, \quad s = 1, \dots, 2S,$$

where $\sum_{s=1}^{2S} p_d^{(s,t)} = 1$.

3. Approximate $\hat{G}_d^{bp} \approx \frac{1}{T} \sum_{t=1}^T G_d^{(t)}$, where $G_d^{(t)} = G_d(u_d^{(t)}, \beta)$, $t = 1, \dots, T$.

Appendix D presents an extensive literature on the MC, QMC and GHQ approximations to a multiple integral in $\mathbb{R}^m$. In particular, some of the references books and paper deal with the estimation of the error and the order of convergence. Concerning the approximation of (7), we remind that $f_\psi(u_d|y_d)$ is multivariate normal. Therefore, considering the condition

$$\int_{\mathbb{R}^m} G_d^2(u_d, \beta) f_\psi(u_d|y_d) du_d < \infty,$$

stochastic convergence in probability of order $kO_p(T^{-1/2})$ is guaranteed. The order of convergence of the GHQ method depends on the regularity conditions on $G_d(u_d, \beta)$ and the dimension of the integral. For univariate integrals, it is possible to obtain deterministic convergence rates of the type $O(T^{-r/2})$, $r \geq 1$. However, the theoretical and empirical results give lower convergence rates when increasing the dimension of the integrals. This is why, we carry out simulation studies to obtain information about the errors of the approximations to the multiple integral (7).

4. Mean squared error estimation

This section presents two parametric bootstrap algorithms for estimating the MSE of the EBP. For this, we build upon contributions provided by [González-Manteiga et al. \(2008, 2010\)](#).

4.1. Parametric bootstrap for MSE

For estimating the MSE of $\hat{G}_d^{ebp}$, $MSE(\hat{G}_d^{ebp}) = E[(\hat{G}_d^{ebp} - G_d)^2]$, we propose a parametric bootstrap estimator. We calculate the REML estimators $\hat{\beta}$ and $\hat{\theta}$ by using the observed data (y_d, X_d) , $d = 1, \dots, D$, and run the following algorithm.

Algorithm B.

1. Repeat B times ($b = 1, \dots, B$)

(a) For $d = 1, \dots, D$, generate $u_d^{*(b)} \sim N_m(0, \hat{V}_{ud})$, $e_d^{*(b)} \sim N_m(0, V_{ed})$ and calculate

$$y_d^{*(b)} = \mu_d^{*(b)} + e_d^{*(b)}, \quad \mu_d^{*(b)} = X_d \hat{\beta} + u_d^{*(b)}, \quad G_d^{*(b)} = g(\mu_{d1}^{*(b)}, \dots, \mu_{dm}^{*(b)}). \quad (13)$$

- (b) By using the bootstrap data $(y_d^{*(b)}, X_d)$, $d = 1, \dots, D$, calculate
- REML estimators $\hat{\beta}^{*(b)}$ and $\hat{\theta}^{*(b)}$.
 - EBPs $\hat{G}_d^{ebp*(b)}$ by MC integration or GHQ under the MFH model (13).

2. For $d = 1, \dots, D$, calculate $mse_{1d}^* = \frac{1}{B} \sum_{b=1}^B (\hat{G}_d^{ebp*(b)} - G_d^{*(b)})^2$.

4.2. Parametric bootstrap for a component of the MSE

The MSE contains integrals with respect to the joint p.d.f. $f(y_d, u_d) = f(u_d|y_d)f(y_d)$. Note that

$$G_d = G_d(u_d, \beta), \quad \hat{G}^{bp} = \hat{G}^{bp}(y_d, \beta, \theta), \quad \hat{G}^{ebp} = \hat{G}^{bp}(y_d, \hat{\beta}, \hat{\theta}) = \hat{G}^{ebp}(y_d).$$

The MSE of the $\hat{G}_d^{ebp}$ is

$$\begin{aligned} E[(\hat{G}_d^{ebp} - G_d)^2] &= E[(\hat{G}_d^{ebp} - \hat{G}_d^{bp}) + (\hat{G}_d^{bp} - G_d)]^2 \\ &= E[(\hat{G}_d^{ebp} - \hat{G}_d^{bp})^2] + E[(\hat{G}_d^{bp} - G_d)^2] \triangleq S_{1d} + S_{2d}. \end{aligned}$$

As $E[G_d|y_d] = \hat{G}_d^{bp}$, the cross-moment term is

$$E[(\hat{G}_d^{ebp} - \hat{G}_d^{bp})(\hat{G}_d^{bp} - G_d)] = E_{y_d}[(\hat{G}_d^{ebp} - \hat{G}_d^{bp})E[\hat{G}_d^{bp} - G_d|y_d]] = 0.$$

For estimating $S_{1d} = E[(\hat{G}_d^{ebp} - \hat{G}_d^{bp})^2]$, we use the parametric bootstrap algorithm. We calculate the REML estimators $\hat{\beta}$ and $\hat{\theta}$ by using the observed data (y_d, X_d) , $d = 1, \dots, D$, and run the following algorithm.

Algorithm S1.

- Repeat B times ($b = 1, \dots, B$)
 - For $d = 1, \dots, D$, generate $u_d^{*(b)} \sim N_m(0, \hat{V}_{ud})$, $e_d^{*(b)} \sim N_m(0, V_{ed})$ and calculate

$$y_d^{*(b)} = \mu_d^{*(b)} + e_d^{*(b)}, \quad \mu_d^{*(b)} = X_d \hat{\beta} + u_d^{*(b)}. \quad (14)$$

- By using the bootstrap data $(y_d^{*(b)}, X_d)$, $d = 1, \dots, D$, calculate
 - REML(ML) estimators $\hat{\beta}^{*(b)}$ and $\hat{\theta}^{*(b)}$.
 - BPs $\hat{G}_d^{bp*(b)}$ and EBPs $\hat{G}_d^{ebp*(b)}$ by MC integration or GHQ under model (14).
- For $d = 1, \dots, D$, calculate $\hat{S}_{1d}^* = \frac{1}{B} \sum_{b=1}^B (\hat{G}_d^{ebp*(b)} - \hat{G}_d^{bp*(b)})^2$.

The second summand is

$$S_{2d} = E[(\hat{G}_d^{bp} - G_d)^2] = E_{y_d}[E[(G_d - \hat{G}_d^{bp})^2|y_d]] = E_{y_d}[\text{var}[G_d|y_d]].$$

In integral form, we have

$$S_{2d} = \int_{\mathbb{R}^m} \underbrace{\left\{ \int_{\mathbb{R}^m} (\hat{G}_d^{bp} - G_d)^2 f(u_d|y_d) du_d \right\}}_{S_{2d}^{in}} f(y_d) dy_d. \quad (15)$$

We calculate the REML estimators $\hat{\beta}$ and $\hat{\theta}$ by using the observed data (y_d, X_d) , $d = 1, \dots, D$, and run the following algorithm to approximate S_{2d} via GHQ. We use a nested GHQ algorithm for the nested integrals. The algorithm is derived for known parameters β and θ and estimated by plugging in $\hat{\theta}$ and $\hat{\beta}$.

Algorithm S2.1.

- For $d = 1, \dots, D$, do
 - Take decomposition $\hat{V}_d(\hat{\theta}) = P_d P_d'$ and $z_d = 2^{-0.5} P_d^{-1}(y_d - X_d \hat{\beta})$, $y_d = \sqrt{2} P_d z_d + X_d \hat{\beta}$, such that

$$\begin{aligned} S_{2d} &= \int_{\mathbb{R}^m} \left\{ \int_{\mathbb{R}^m} (\hat{G}_d^{bp} - G_d)^2 f(u_d|y_d) du_d \right\} S_{2d}^{in} f(y_d) dy_d \\ &= \pi^{-m/2} \int_{\mathbb{R}^m} S_{2d}^{in} (\sqrt{2} P_d z_d + X_d \hat{\beta}) \exp(-z_d' z_d) dz_d. \end{aligned}$$

(b) Calculate approximation

$$\hat{S}_{2d} = \sum_{t_1=1}^T \cdots \sum_{t_m=1}^T \pi^{-m/2} \hat{S}_{2d}^{in}(\tilde{z}_{t_1, \dots, t_m}) \prod_{k=1}^m w_{t_k},$$

with $\tilde{z}_{t_1, \dots, t_m} = \sqrt{2}P_d(z_{t_1}, \dots, z_{t_m})' + X_d \hat{\beta}$, where w_{t_k} and z_{t_k} are the standard GHQ weights and nodes for degree T , $k = 1, \dots, m$, $t = 1, \dots, T$. The calculation of $\hat{S}_{2d}^{in} = \hat{S}_{2d}^{in}(y_d)$ is described in the following.

- i. For $f_{\hat{\psi}}(u_d|y_d)$ we have $\mu_{u_d|y_d}(\hat{\beta}, \hat{\theta}) = \hat{V}_{ud}(\hat{\theta})\hat{V}_d^{-1}(\hat{\theta})(y_d - X_d \hat{\beta})$ and $\hat{V}_{u_d|y_d}(\hat{\theta}) = \hat{V}_{ud}(\hat{\theta})\hat{V}_d^{-1}(\hat{\theta})V_{ed}$.
- ii. Take decomposition $\hat{V}_{u_d|y_d}(\hat{\theta}) = L_d L_d'$ and $z_d = 2^{-0.5} L_d^{-1}(u_d - \mu_{u_d|y_d})$, $u_d = \sqrt{2}L_d z_d + \mu_{u_d|y_d}$, such that

$$\hat{G}_d^{bp} = \pi^{-m/2} \int_{\mathbb{R}^m} G_d(\sqrt{2}L_d z_d + \mu_{u_d|y_d}, \hat{\beta}) \exp\left\{-z_d' z_d\right\} dz_d.$$

iii. Calculate approximation

$$\begin{aligned} \hat{S}_{2d}^{in} &= \sum_{t_1=1}^T \cdots \sum_{t_m=1}^T \pi^{-m/2} (G_d(\tilde{z}_{t_1, \dots, t_m}) - \hat{G}_d^{bp})^2 \prod_{k=1}^m w_{t_k}, \\ \hat{G}_d^{ebp} &\approx \sum_{t_1=1}^T \cdots \sum_{t_m=1}^T \pi^{-m/2} (G_d(\tilde{z}_{t_1, \dots, t_m}))^2 \prod_{k=1}^m w_{t_k}, \end{aligned}$$

with $\tilde{z}_{t_1, \dots, t_m} = \sqrt{2}L_d(z_{t_1}, \dots, z_{t_m})' + \mu_{u_d|y_d}$, where w_{t_k} and z_{t_k} are the standard GHQ weights and nodes for degree T , $k = 1, \dots, m$, $t = 1, \dots, T$.

Finally, an estimator of the MSE of $\hat{G}_d^{ebp}$ is

$$mse_{2d} = \hat{S}_{1d}^* + \hat{S}_{2d}.$$

As estimator of S_{2d} , we further propose an estimator of the inner integral. This is to say, we avoid approximating the double integral by estimating an expected value (expectation with respect to $f(y_d)$) with observed values. Avoiding to approximate the double integral can be especially useful for higher dimensional problems. The inner integral is

$$S_{2d}^{in} = \text{var}[G_d|y_d] = \int_{\mathbb{R}^m} (G_d - \hat{G}_d^{bp})^2 f(u_d|y_d) du_d, \quad \hat{G}_d^{bp} = E[G_d|y_d].$$

Algorithm S2.2.

We use the procedure for calculating approximation $\hat{S}_{2d}^{in}$ in Algorithm S2.1. with observed y_d and set $\hat{S}_{2d}^{in} = \hat{S}_{2d}^{in}(y_d)$. To put it differently, we only apply the inner GHQ of Algorithm S2.1 with observed y_d . Finally, an estimator of the MSE of $\hat{G}_d^{ebp}$ is

$$mse_{3d}^* = \hat{S}_{1d}^* + \hat{S}_{2d}^{in}.$$

5. Simulations

This section presents some simulation experiments to analyze the new predictors of unemployment rates and the corresponding MSE estimators. See Appendix C for details about EBP calculation of unemployment rates. The unemployment rate represents a rather simple non-linear function and we expect the bias of the plug-in predictors for this function to be small. Thus, if the EBPs show better performance than plug-in predictors for this simple non-linear function we expect this performance improvement for more complicated non-linear functions in particular. For the unemployment rate, let $m = 2$ and the domain parameters of interest take the form

$$R_d = R_d(u_d, \beta) = \frac{\mu_{d1}}{\mu_{d1} + \mu_{d2}} = \frac{x_{d1}'\beta_1 + u_{d1}}{x_{d1}'\beta_1 + u_{d1} + x_{d2}'\beta_2 + u_{d2}}, \quad d = 1, \dots, D.$$

We employ the REML fitting algorithm described in Appendix B. Let us consider the particular bivariate Fay- Herriot (BFH) model

$$y_d = X_d \beta + u_d + e_d, \quad d = 1, \dots, D, \quad (16)$$

with $p_1 = p_2 = 2$, $p = 4$, $\beta_1 = (\beta_{11}, \beta_{12})' = (0, 0.2)'$, $\beta_2 = (\beta_{21}, \beta_{22})' = (0.6, 0.2)'$. For $k = 1, 2$, $d = 1, \dots, D$, we generate X_d before starting the simulation loop and it remains fixed. Do $X_d = \text{diag}(x_{d1}, x_{d2})_{2 \times 4}$, $x_{d1} = (x_{d11}, x_{d12})$, $x_{d2} = (x_{d21}, x_{d22})$, $x_{d11} = x_{d21} = 1$ and $x_{d12} = U_{d1}$, $U_{d1} \stackrel{\text{ind}}{\sim} \text{Uniform}(0.49, 0.51)$, $x_{d22} = U_{d2}$, $U_{d2} \stackrel{\text{ind}}{\sim} \text{Uniform}(0.49, 0.51)$. For $d = 1, \dots, D$, $u_d \sim N_2(0, V_{ud})$ and $e_d \sim N_2(0, V_{ed})$ are simulated in each iteration. We take

$$V_{ud} = \begin{pmatrix} \sigma_{u1}^2 & \rho_{u1}\sigma_{u1}\sigma_{u2} \\ \rho_{u1}\sigma_{u1}\sigma_{u2} & \sigma_{u2}^2 \end{pmatrix}, \quad V_{ed} = \begin{pmatrix} \sigma_{e1}^2 & \rho_{e1}\sigma_{e1}\sigma_{e2} \\ \rho_{e1}\sigma_{e1}\sigma_{e2} & \sigma_{e2}^2 \end{pmatrix},$$

Table 1Statistics of BPs for varying number of function evaluations E (in %).

Statistic	BP	Algorithm	$E = 16$	24/25	100	400	2500	10000	40000
$RBIAS_d$	IN-FH		0.026	0.026	0.026	0.026	0.026	0.026	0.026
	IN-BFH		0.026	0.026	0.026	0.026	0.026	0.026	0.026
	I1	1_{mc}	-0.102	-0.101	-0.079	0.011	-0.013	-0.012	-0.013
		1_{ghq}	-0.015	-0.013	-0.013	-0.013	-0.013	-0.013	-0.013
	I1 $_{\bullet}$	1_{mc}	0.496	0.951	-0.288	0.388	0.035	-0.001	-0.015
		1_{ghq}	-0.038	-0.008	-0.013	-0.013	-0.013	-0.013	-0.013
	I2	2_{mc}	-0.012	-0.012	-0.013	-0.014	-0.013	-0.013	-0.013
		2_{ghq}	-0.013	-0.013	-0.013	-0.013	-0.013	-0.013	-0.013
	I3	$3_{mc}, S = 5$	0.135	0.135	0.090	-0.034	-0.021	-0.016	-0.014
	IN-FH		8.834	8.834	8.834	8.834	8.834	8.834	8.834
	IN-BFH		8.834	8.834	8.834	8.834	8.834	8.834	8.834
	I1	1_{mc}	9.388	9.168	8.953	8.823	8.843	8.837	8.837
		1_{ghq}	8.848	8.832	8.834	8.834	8.834	8.834	8.834
$RRMSE_d$	I1 $_{\bullet}$	1_{mc}	38.930	29.711	13.980	10.988	9.159	8.892	8.863
		1_{ghq}	8.865	8.830	8.834	8.834	8.834	8.834	8.834
	I2	2_{mc}	8.833	8.833	8.833	8.834	8.834	8.834	8.834
		2_{ghq}	8.834	8.834	8.834	8.834	8.834	8.834	8.834
	I3	$3_{mc}, S = 5$	10.693	10.693	9.368	9.073	8.940	8.943	8.937

with $\sigma_{u1} = \sigma_{u2} = 0.012$, $\sigma_{e1} = \sigma_{e2} = 0.014$. Note that the parametrization is chosen such that the simulations yield realistic unemployment rates when compared to empirical practice. We used the Spanish labour force survey (SLFS) for reference. Recall that $\theta_1 = \sigma_{u1}^2$, $\theta_2 = \sigma_{u2}^2$, and $\theta_3 = \rho_{12} = \rho_u$. We consider $\rho_u = \rho_e = -0.5$ as a realistic scenario, i.e. both the random effects and sampling errors of domain specific employment and unemployment proportions are negatively correlated. The above parameter configurations generate y_{1d} and y_{2d} as the proportions (not totals) of unemployed and employed in domain d . The simulation experiments generate proportions of unemployed, employed and inactive persons around 0.1, 0.7 and 0.2, respectively. This implies unemployment rates around 0.125 (12.5%).

5.1. Simulation 1: Approximations of BP

Simulation 1 evaluates the different procedures presented in Section 3 for approximating unemployment rates, i.e. the plug-in prediction and MC and GHQ approximations to the integral forms 1, 1 $_{\bullet}$, 2, and 3. The focus of this first simulation is purely on the comparison of these approximations. We therefore choose $D = 1$ to focus on a single area and set parameters $\psi = (\beta', \theta')'$ to their true values. We consider the case of $\rho_u = \rho_e = -0.5$. For performance measurement, we consider the relative bias (RBIAS) and relative root mean squared error (RRMSE). They are calculated as averages across domains and set into a percentage relative to the true domain unemployment rates. A detailed description and exact formulas of performance measures for Simulation 1 are provided in Appendix E.1 (supplemental material). The only randomness considered in this simulation is from generating the random effects u_d and sampling errors e_d in each iteration. For GHQ method we apply eigenvalue decomposition of the covariance matrix. To have a fair comparison of MC, SIR, and GHQ methods, described in Section 3, we evaluate them for the same number of function evaluations $E = 100$.

1. For GHQ, we take degree $T = 10$. By product rule, the number of function evaluations is $E = T^2 = 100$.
2. For MC approximation to integrals I1, I1 $_{\bullet}$, and I2, we take $T = 50$. Due to the use of a antithetic variates, the number of function evaluations is $E = 2T = 100$.
3. For SIR approximation to I3 when $S = 5$ we take $T = 10$. Due to the use of a antithetic variates, the number of function evaluations is $E = 2ST = 100$.

For the IN-FH predictor $\hat{K}_d^{fh,in}$ and the IN-BFH predictor $\hat{K}_d^{in}$, there is no need to approximate integrals. Therefore, we only report one value of the performance measures in Table 1 and repeat them over variations of E as reference values. For these predictors, the relative biases ($RBIAS_d$ in %) are 0.026 and 0.026 respectively and the relative root MSEs ($RRMSE_d$ in %) are 8.834 and 8.834 respectively. This is to say, the IN-BFH predictor is not taken advantage of the bivariate modelling in the simulated scenario $\rho_u = \rho_e = -0.5$. As both correlations ρ_u and ρ_e are of same magnitude and same sign, we do not expect large gains by multivariate modeling. See Datta et al. (1999) who show that the gain by applying the bivariate instead of univariate FH estimator is highest when correlations ρ_u and ρ_e are of opposite sign and high magnitude.

Table 1 presents the simulation results for the BPs. For GHQ we only use squared numbers of integers because of the product rule, whereas for MC we only consider even numbers of function evaluations due to the use of the a antithetic variates. The case (column) of 24/25 function evaluations therefore corresponds to 25 evaluations for GHQ and 24 evaluations for MC.

From Table 1, we give the following conclusions. First, applying the proposed approximations to the BP instead of the plug-in estimates give lower $RBIAS_d$ and similar (or lower) $RRMSE_d$ and should therefore be preferred. Second, comparing the approximations of the different forms of the integral of the BP, we prefer I2 as it shows a good performance both for

Table 2Statistics for $\rho_e = -0.5$ (in %).

Statistic	θ, β	Predictor	$\rho_u = -0.75$	-0.5	-0.25	0	0.25	0.5	0.75
RBIAS	True	IN-FH	0.021	0.032	0.076	0.071	0.110	0.146	0.197
		IN-BFH	0.035	0.032	0.058	0.034	0.039	0.065	0.069
		BP-GHQ	0.010	-0.007	0.010	-0.021	-0.021	0.002	0.003
	REML	IN-FH	0.024	0.027	0.085	0.083	0.108	0.157	0.187
		IN-BFH	0.037	0.027	0.066	0.044	0.046	0.072	0.077
		EBP-GHQ	0.014	-0.012	0.019	-0.010	-0.013	0.008	0.011
ARBIAS	True	IN-FH	0.157	0.144	0.164	0.151	0.166	0.191	0.215
		IN-BFH	0.158	0.144	0.157	0.140	0.132	0.136	0.109
		BP-GHQ	0.158	0.141	0.147	0.139	0.130	0.128	0.094
	REML	IN-FH	0.157	0.146	0.168	0.158	0.168	0.198	0.208
		IN-BFH	0.157	0.148	0.160	0.147	0.133	0.138	0.116
		EBP-GHQ	0.156	0.145	0.148	0.142	0.130	0.128	0.100
RRMSE	True	IN-FH	8.767	8.608	8.455	8.305	8.154	8.000	7.834
		IN-BFH	8.690	8.608	8.393	8.067	7.597	6.954	6.016
		BP-GHQ	8.690	8.608	8.392	8.067	7.597	6.954	6.016
	REML	IN-FH	8.877	8.715	8.567	8.427	8.269	8.122	7.962
		IN-BFH	8.826	8.744	8.539	8.237	7.777	7.171	6.279
		EBP-GHQ	8.826	8.744	8.539	8.237	7.777	7.171	6.279

Table 3Statistics for $\rho_e = 0$ (in %).

Statistic	θ, β	Predictor	$\rho_u = -0.75$	-0.5	-0.25	0	0.25	0.5	0.75
RBIAS	True	IN-FH	-0.006	0.015	0.041	0.088	0.131	0.136	0.156
		IN-BFH	0.019	0.043	0.055	0.088	0.114	0.100	0.099
		BP-GHQ	-0.010	-0.007	-0.011	0.010	0.025	0.004	-0.004
	REML	IN-FH	-0.031	0.009	0.036	0.092	0.121	0.130	0.160
		IN-BFH	0.001	0.034	0.051	0.091	0.103	0.093	0.103
		EBP-GHQ	-0.026	-0.013	-0.012	0.014	0.016	-0.003	0.001
ARBIAS	True	IN-FH	0.152	0.155	0.150	0.154	0.175	0.183	0.190
		IN-BFH	0.138	0.153	0.153	0.154	0.165	0.163	0.148
		BP-GHQ	0.140	0.145	0.145	0.136	0.134	0.136	0.124
	REML	IN-FH	0.155	0.159	0.155	0.156	0.170	0.182	0.194
		IN-BFH	0.141	0.157	0.159	0.157	0.159	0.162	0.153
		EBP-GHQ	0.142	0.152	0.151	0.138	0.134	0.140	0.123
RRMSE	True	IN-FH	8.527	8.379	8.234	8.071	7.918	7.758	7.581
		IN-BFH	7.978	8.141	8.173	8.071	7.851	7.459	6.827
		BP-GHQ	7.978	8.141	8.173	8.071	7.850	7.458	6.827
	REML	IN-FH	8.619	8.482	8.337	8.181	8.034	7.877	7.704
		IN-BFH	8.108	8.279	8.305	8.209	8.004	7.627	7.024
		EBP-GHQ	8.108	8.279	8.305	8.209	8.004	7.626	7.024

MC integration and GHQ with relatively few function evaluations. Third, GHQ is superior to MC integration especially for a small number of evaluation points. Furthermore, the results of the GHQ are less volatile already for a small number of function evaluations. As long as the function over which we integrate can be well approximated by a polynomial of chosen degree T , and we consider problems of low dimension like $m = 2$, GHQ is the approximation method of choice. Therefore, in Simulation 2 and 3 for the approximation of BP of the unemployment rate we choose GHQ and I2. Appendix E.2 presents additional results for $\rho_u = -0.5$ with $\rho_e = 0$ and $\rho_e = 0.25$ respectively.

5.2. Simulation 2: EBP and plug-in predictors

Simulation 2 evaluates the performance of the plug-in predictors and approximations to the (E)BP of the unemployment rates based on the univariate and bivariate Fay-Herriot model. Next to the evaluation of the predictors with REML parameter estimates, it also considers the predictors calculated with true model parameters as reference. We take $D = 200$ domains and correlations ρ_u and ρ_e with values in the set $(-0.75, -0.5, -0.25, 0, 0.25, 0.5, 0.75)$. For the approximation of the BP and EBP of the unemployment rate we use GHQ with degree $T = 5$ in each dimension and thus a total of 25 evaluated points. Gauss-Hermite quadrature is applied to integral form I2 as the combination showed good performance in Simulation 1. As in Simulation 1, we consider the RBIAS and RRMSE averaged across domains as performance measures. Moreover, we use the absolute relative bias (ARBIAS) to consider the overall level of deviation in the predictions. Like RBIAS and RRMSE, it is averaged across domains and set into a percentage relative to the true domain unemployment rates. The details of Simulation 2 including all formulas are described in the Appendix E.3.

Tables 2 and 3 present the simulation results for two cases, under true and estimated parameters, and different correlation structures. Considering the plug-in predictors, the BFH plug-in is superior to the univariate FH plug-in in terms of

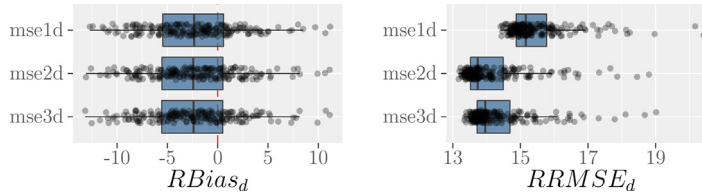


Fig. 1. Performance of MSE estimators (in %): $RBias_d$ and $RRMSE_d$.

Table 4

Performance measures of MSE estimators (in %).

MSE estimator	RBIAS	ARBIAS	RRMSE
mse_{1d}^*	-2.392	4.101	15.467
mse_{2d}^*	-2.386	4.107	14.140
mse_{3d}^*	-2.387	4.105	14.336

ARBIAS and RRMSE. With respect to the RBIAS, we see that the BFH plug-in is less biased for the majority of settings under $\rho_e = -0.5$, and for positive random effect correlation under $\rho_e = 0$. The BP and EBP outperform both plug-in predictors in almost all scenarios in terms of all bias-related measures. With respect to the RRMSE, their performance is very close to the BHF plug-in. Please note that this is in line with the theoretical developments presented in Section 3. The BP is constructed such that it has minimal MSE within the class of all model-unbiased predictors. The BHF plug-in is biased, which is why theory cannot guarantee the BP's MSE-superiority in this setting. However, especially when investigating the RBIAS, it is evident the bias reduction achieved by the BP and EBP is considerable. Furthermore, concerning the computational cost, we calculate a sufficiently good GHQ approximation of the BP with a rather small number of function evaluations. In addition, Appendix E.4 presents similar results for a example of a trivariate case, i.e. with three dependent variables.

5.3. Simulation 3: MSE estimators

Simulation 3 investigates the performance of the bootstrap estimators of the MSEs of the EBPs of unemployment rates. It also gives some advices about the number of bootstrap replicates for the MSE estimation. We recall that the bootstrap-based estimators are mse_{1d}^* and S_{1d}^* , defined in Sections 4.1 and 4.2 respectively. For the integral form I2, we apply GHQ with $T = 5$ degrees, i.e. a total of 25 function evaluations and use eigenvalue decomposition of the covariance matrix. The MSE estimators are described in Section 4. As in Simulation 1, we take $\rho_u = \rho_e = -0.5$ and evaluate the MSE estimators for $D = 200$ areas.

A first simulation experiment evaluates how many bootstrap replicates are needed for the MSE estimation. For this, we simulate the data once, estimate the coefficients of the BFH model, and apply $B = 1000$ bootstrap replicates. This preliminary simulation is described in Appendix E.5. The result is that for calculating mse_{1d}^* we can implement the bootstrap algorithm with $B = 400$ replicates, so that we will obtain a similar precision to $B = 1000$ replicates. For $\hat{S}_{1d}^*$ the bootstrap results are more volatile and we recommend $B = 500$ replicates.

For a joint simulation study of the MSE estimators we consider $B = 500$ bootstrap replicates. In Appendix E.6 the simulation study concerning the performance of the proposed MSE estimators is described in more detail. Figure 1 shows the resulting relative bias (RBIAS) and RRMSE (in %) of the proposed estimators over all domains as percentage relation of the respective true domain MSE of prediction. The mean values over all domains are presented in Table 4. First, we see that although there is a slight underestimation of the true MSE, the RBias of the estimators is small and they are therefore considered suitable. The measures for mse_{2d}^* and mse_{3d}^* are close, implying that S_{2d}^{in} is a good estimator of S_{2d} which can be especially useful for higher-dimensional problems as it reduces the number of function evaluations considerably. In terms of RRMSE the proposed separate estimation of the two MSE summands, as done in mse_{2d}^* and mse_{3d}^* , is superior to the solely parametric bootstrap estimation of the MSE given by mse_{1d}^* .

6. Application to Spanish labour force data

As an illustrative example, we apply the introduced methodology to the estimation of small area unemployment rates by using data from the Spanish labour force survey (SLFS) in the first quarter of 2021. The SLFS studies the Spanish labour market, evaluates the number of employed, unemployed and inactive persons that are 16 years old or more, and analyzes socio-economic characteristics of demographic groups defined by sex, age, level of education, nationality and others. The SLFS is harmonized to the standards of the International Labour Organization (ILO) and the European Statistical Office (EUROSTAT) such that their results are comparable to those published by other European countries.

The SLFS is a quarterly survey carried out by the Spanish Statistical Office (INE). The samples are extracted by two-stage stratified random sampling from each Spanish province. The Primary Sampling Units are municipality areas with around

Table 5

Quantiles of the estimated correlations $\hat{\rho}_{ed}$,
 $d = 1, \dots, D$.

0%	25%	50%	75%	100%
-0.58	-0.38	-0.24	-0.15	-0.04

3000 persons, which are grouped into strata according to the size of the municipality. The Secondary Sampling Units are dwellings. The final units are the individuals aged 16 or more in the selected dwelling.

The micro data from the SLFS is free downloadable from the INE web site. For confidentiality reasons, the geographical variables below the level of province are not included. The sample data contains 91 variables giving information about location, demography, education, studies, labour activity during the reference week, employment characteristics, search of employment and work absenteeism, among others. The SLFS data set also contains the elevation factors, which are employed to calculate direct estimators by provinces. The elevation factors are calculated from the inverses of the inclusion probabilities after correction by non-response and calibration to known province quantities.

In the SLFS data the persons' age is given in 5-year intervals. From that we define broader age classes AGE1: 16-24, AGE2: 25-54, and AGE3: 55-64. As domains of interest we take the cross-combinations of three age classes (AGE1, AGE2, AGE3), sex and 52 provinces, including the autonomous cities of Ceuta and Melilla. As variables of interest in the BFH model, we take the proportions of unemployed and employed in the domains. There is one domain (province 49 - Zamora, male, AGE1) with no unemployed person, thereby the direct estimate of unemployed is zero and no variance estimate can be calculated. We exclude this domain and thereby consider the remaining $D = 311$ domains as domains interest. We partition sample s into domain-specific samples s_d , $d = 1, \dots, D$. Let y_{dij} be the binary variable (1 if yes and 0 if no) referring to the status unemployed ($i = 1$) and employed ($i = 2$), for individual j from domain d . Let w_{dj} be the non-negative elevation factor for individual j in domain d . The direct estimator of the number of unemployed and employed is calculated by

$$\hat{Y}_{di}^{dir} = \sum_{j \in s_d} w_{dj} y_{dij}, \quad d = 1, \dots, D, \quad i = 1, 2.$$

Domain sizes are not free available and therefore estimated by

$$\hat{N}_d = \sum_{j \in s_d} w_{dj}, \quad d = 1, \dots, D.$$

As no second-order inclusion probabilities are provided, we approximate the design-based variance of $\hat{Y}_{di}^{dir}$ by using the Hajek-approximation depending only on first-order inclusion probabilities [Hájek \(1964\)](#). The design-based covariances of two direct total estimators are approximated by [Wood \(2008\)](#)

$$\widehat{\text{cov}}_{\pi}(\hat{Y}_{d1}^{dir}, \hat{Y}_{d2}^{dir}) = \frac{1}{2} \left(\widehat{\text{var}}_{\pi}(\hat{Y}_{d1}^{dir}) + \widehat{\text{var}}_{\pi}(\hat{Y}_{d2}^{dir}) - \widehat{\text{var}}_{\pi}(\hat{Y}_{d1}^{dir} - \hat{Y}_{d2}^{dir}) \right), \quad d = 1, \dots, D.$$

Direct estimators of proportions of unemployed and employed, with their variance and covariance estimators, are calculated by dividing the total estimators by $\hat{N}_d$ and their variances and covariances by $\hat{N}_d^2$. This is to say, $\hat{Y}_{di}^{dir} \triangleq \hat{Y}_{di}^{dir} / \hat{N}_d$ and

$$\sigma_{edi}^2 \triangleq \widehat{\text{var}}_{\pi}(\hat{Y}_{di}^{dir}) = \frac{1}{\hat{N}_d^2} \widehat{\text{var}}_{\pi}(\hat{Y}_{di}^{dir}), \quad d = 1, \dots, D, \quad i = 1, 2,$$

$$\widehat{\text{cov}}_{\pi}(\hat{Y}_{d1}^{dir}, \hat{Y}_{d2}^{dir}) = \frac{1}{\hat{N}_d^2} \widehat{\text{cov}}_{\pi}(\hat{Y}_{d1}^{dir}, \hat{Y}_{d2}^{dir}), \quad d = 1, \dots, D.$$

where the symbol $\triangleq$ defines components of the multivariate FH models to be selected, in the notation of [Section 2](#). The estimated design-based correlations between the direct estimators of the proportions of unemployed and employed are given by

$$\hat{\rho}_{ed} \triangleq \widehat{\text{cov}}_{\pi}(\hat{Y}_{d1}^{dir}, \hat{Y}_{d2}^{dir}) / \left(\widehat{\text{var}}_{\pi}(\hat{Y}_{d1}^{dir}) \widehat{\text{var}}_{\pi}(\hat{Y}_{d2}^{dir}) \right)^{1/2}, \quad d = 1, \dots, D.$$

Simulation 2 ([Section 5.2](#)) shows that the correlations of the sampling errors of the dependent variables plays an important role in multivariate FH models. [Table 5](#) shows that the sampling errors, e_{d1} and e_{d2} , of the direct estimators, $\hat{Y}_{d1}^{dir}$ and $\hat{Y}_{d2}^{dir}$, have a negative correlation of moderate to small magnitude.

To select a bivariate Fay-Herriot model, suitable auxiliary variables are needed. According to the theoretical developments presented in [Section 2](#), these should be accurate (non-estimated) values obtained from a data source that is independent from the target variable survey. Unfortunately, for the domains of interest we could not find suitable publicly available auxiliary information. As a compromise, we used all four SLFS waves from 2020 to calculate direct estimators of the required covariate records. The auxiliary information are not exact values, but sample estimates that are subject to errors. Hence, a measurement error model as proposed by [Ybarra and Lohr \(2008\)](#), [Arima et al. \(2017\)](#), and [Burgard et al. \(2020, 2021, 2022a\)](#) should be applied. However, this is beyond the scope of this article and a subject for future research. Note that by

Table 6

Variance component estimates and asymptotic 95% confidence intervals.

	$\hat{\theta}$	Lower limit	Upper limit
$\hat{\sigma}_{u1}$	0.028	0.024	0.031
$\hat{\sigma}_{u2}$	0.052	0.046	0.058
$\hat{\rho}_{12}$	-0.731	-0.832	-0.629

Table 7

Parameter estimates for proportion of unemployed and employed.

	Unemployed				Employed			
	$\hat{\beta}$	std.error	t-value	p-value	$\hat{\beta}$	std.error	t-value	p-value
(Intercept)	0.025	0.028	0.915	0.360	0.404	0.051	7.903	0.000
Edu.Pri	0.231	0.037	6.308	0.000	-0.611	0.067	-9.094	0.000
Edu.Sec	0.102	0.034	3.019	0.003	-0.319	0.062	-5.121	0.000
AGE2	0.038	0.011	3.425	0.001	0.474	0.020	24.078	0.000
AGE3	-0.035	0.009	-3.978	0.000	0.362	0.016	22.534	0.000
sex.male	-0.015	0.004	-3.428	0.001	0.122	0.008	15.469	0.000

using the four waves of 2020, the auxiliary information are based on four times as many observations as the direct target variable estimators. That is to say, we can significantly increase the effective sample size for each domain and thus improve the estimation results despite sampling errors on the covariate records. Accordingly, for an illustrative example, this is still a suitable solution to demonstrate the effectiveness of the methodology.

The model covariates are derived from the education variables primary or less (Edu.Pri) and secondary (Edu.Sec). We also consider the nationality variable Spanish (Nat.Spa). They are all binary variables at the unit-level. The superior education and non-Spanish variables (reference categories) are not included in the set of explanatory variables. The candidates to become model covariates are the direct estimators of the proportion of persons within the group defined by each auxiliary variable: Edu.Pri, Edu.Sec, and Nat.Spa.

A BFH model is fitted to the proportions of unemployed and employed as the dependent variables. The corresponding estimated variances and covariances are used to form V_{ed} , $d = 1, \dots, D$. After considering different models, for both dependent variables the final auxiliaries are Edu.Pri, Edu.Sec and the dummy variables AGE2, AGE3, and male sex. For estimating the model parameters, we apply the REML Fisher-Scoring algorithm as we did in the simulation studies.

We first discuss the output of the bivariate Fay-Herriot model. Table 6 displays the estimated variance components with 95% confidence intervals. None of the confidence intervals of the random effects variances contains the zero, so that the random effects explain part of the variability of the target variables. As expected, the correlation of the random effects of the proportions of unemployed and employed is considerably negative, also considering the confidence interval.

Table 7 gives the estimated regression parameters. We note that all parameter are significant, i.e. their associated p -values are < 0.05 . The reference category for education is superior education. Therefore, it makes sense that a higher (lower) proportion of persons in superior education is associated with a higher (lower) proportion of employed and a lower (higher) proportion of unemployed. We can see that from the positive and negative coefficients for unemployment and employment respectively. Furthermore, we see from the estimated coefficients that domains for males are associated with higher employment and lower unemployment proportions. As this is an illustrative example based on estimated covariate records, the coefficients values should be treated with caution.

To check the suitability of the model several model diagnostics are presented in Figure 2 for the proportion of unemployed (left) and employed (right). As proposed in Brown et al. (2001), the direct estimates are plotted against the EBLUPs in row one and the diagonal line $y = x$ is added to the plots. From theory, we expect the EBLUPs to be randomly distributed around the direct estimates. For unemployment we see some deviation to the direct estimates for higher magnitudes of unemployment proportions. From the data analysis, we note that high direct estimates of unemployment are generally associated with high sampling variances of the direct estimators. In those cases, the model does not rely on direct estimates and produces a smoothing effect by providing not-so-extreme EBLUPs. Therefore, we would expect the EBLUPs to deviate from the direct estimates especially for high values of unemployment as is visible in row one of Figure 2. For employment, no systematic deviation from direct estimates is visible.

Next, we take a look at standardized residuals in row two of Figure 2. Given the residuals $r_{dk}^{eblup} = y_{dk} - \hat{\mu}_{dk}^{eblup}$ we calculate the standardized residuals $(r_{dk}^{eblup} - \frac{1}{D} \sum_{d=1}^D r_{dk}^{eblup}) / \text{std}(r_{dk}^{eblup})$, where $\text{std}(r_{dk}^{eblup})$ is the standard deviation of the set of residuals, $d = 1, \dots, D$, $k = 1, 2$. By the definition of the BFH model, the residuals should be normally distributed with mean zero. Furthermore, no systematic differences should be visible in the residual distribution for changing magnitude of the EBLUPs, which is the case and can be seen in row two of Figure 2. It is apparent that for both employment and unemployment most residuals are within range, but there are some outliers for which the absolute values of the standardized residuals are greater 3. There are 1.3% (4/311) values with absolute values larger than 3 for each, the proportion of un-

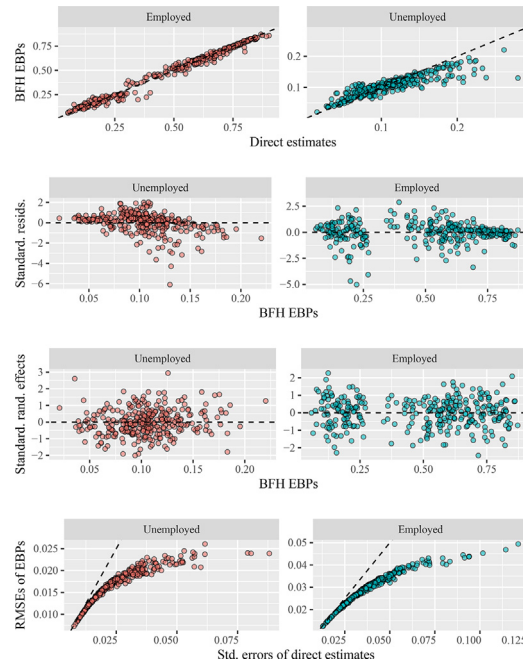


Fig. 2. Model diagnostics for unemployment (left) and employment (right).

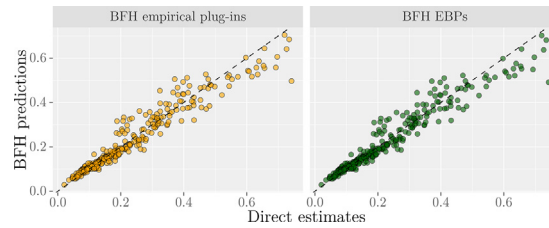


Fig. 3. Direct estimates of unemployment rates versus model-based predictions.

employed and employed. We consider this as acceptable and hence see no model violation in it. For the noticeable large standardized residual for unemployment we find that the underlying sample size of the corresponding direct estimate is very small (41 persons) such that the BFH EBLUP is smoothed more towards the model-based estimator. When extending the present model to account for measurement errors, a special focus has to be placed on the effect of measurement uncertainty in the covariates on the estimates.

To see the efficiency gains of the BFH EBLUPs over the direct estimates, their root MSEs (RMSEs) and standard errors are displayed in row three of Figure 2, again with the additional diagonal line $y = x$. Not only are the EBLUPs always more efficient than the direct estimators, the efficiency gain is especially large when the variance of the direct estimators is high as then the BFH EBLUPs put more weight on the estimated model. For future analysis it would be interesting to further investigate additional auxiliary information and whether they reduce the outlier values in the standardized residuals.

For the unemployment rate, both plug-ins and EBP can be calculated from the BFH model as shown in Section 3. Similar to the procedure in Simulation 2 (Section 5.2) the plug-ins are calculated as $\hat{R}_d^{ein} = \hat{\mu}_{d1}^{eblup} / (\hat{\mu}_{d1}^{eblup} + \hat{\mu}_{d2}^{eblup})$ and EBP $\hat{R}_d^{ebp}$ are calculated with GHQ with $T = 5$ evaluations per dimension. From the LFS data we can also calculate the direct estimator of the unemployment rates. Thereby, the EBP and plug-in predictor from the BFH model can be compared to the direct estimator. In Figure 3 the direct estimates are plotted against the EBP and plug-in predictors from the BFH model. Again, the diagonal line $y = x$ is added to the plot. Both EBP and plug-in predictors do not show systematic differences to the direct estimates. In this application, we see only small differences between the EBP and plug-in predictors of the unemployment rate. From Simulation 2 (Section 5.2) we saw that the difference between the plug-in estimator and the proposed EBP is large when the correlations of sampling errors and random effects are high and of opposite magnitude. In this application, both correlations are estimated to be moderately negative, compare Tables 5 and 6. Thereby, only little difference is expected between the proposed EBP and the plug-in predictor.

In Figure 4 the BFH EBP (row one) and corresponding root MSEs (RMSEs) (row two) are displayed for the 52 provinces by sex for AGE1. Age class AGE1 is of particular interest as the Spanish unemployment rate for young persons (below 25

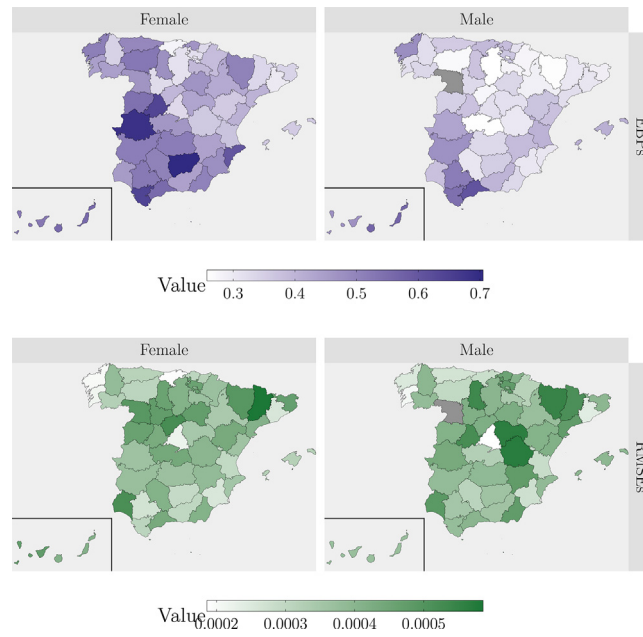


Fig. 4. BFH EBP and their RMSEs for AGE1.

years old) is one of the highest in Europe. As reported by INE, the Spanish unemployment rates for AGE1 were 38% for men and 41% for women during the first trimester of 2021. For AGE2 and AGE3 the corresponding figures are given in Appendix F. As the overall unemployment level can differ to a great extent between the age classes, a unique color scale is used for each plot. For AGE1 the unemployment rate is much higher than for the other age groups. Furthermore, for the statistics on females we observe higher unemployment rates than for the statistics on males. Note that the gray province in Figure 4 (Female) is the province for which the direct estimate of unemployed is missing and therefore not incorporated in the BFH model.

The MSE estimates of the BFH EBP are calculated via $mse_{2d}^* = \hat{S}_{1d}^* + \hat{S}_{2d}$ as the formula showed the lowest RRMSE in Simulation 3 (Section 5.3). Similar to Simulation 3, quantity $\hat{S}_{2d}$ is calculated via the GHQ algorithmic net with $T = 5$, resulting in 625 function evaluations. The resulting MSE estimates are displayed in the second row of Figure 4 for AGE1. The magnitude of the MSEs is quite heterogeneous, indicating that for some domains the model-based predictions worked well whereas for others the estimated unemployment rates should be interpreted with caution. For further research it would be interesting to take a closer look at age class AGE1 and potential additional auxiliary variables to receive more precise predictions.

7. Discussion and conclusions

Small area estimation techniques are widely applied tools to give model-based predictions based on survey information in cases where survey-based direct estimators are associated with high standard errors. Area-level small area models, with the well-known Fay-Herriot model, are applied in cases where either the survey or the covariate information used for modelling are only available at an aggregated level. For several variables of interest, a multivariate Fay-Herriot model can be calculated. Next to the variables themselves, often non-linear combinations of them are of interest. As an example, we consider the unemployment rate which is a non-linear combination of the percentages of employed and unemployed. For the calculation of non-linear combinations, often plug-in estimators based on univariate models, are used, i.e. inserting the single predictions in the non-linear formula. This approach, however, has two drawbacks. First, it does not take the multivariate structure of the variables into account. Second, it produces biased predictors of non-linear multivariable indicators, even in the case of unbiasedly predicting the components.

This paper examines how instead of plug-in predictors, BPs of potentially non-linear combinations of dependent variables can be approximated in MFH models. The BPs have minimum MSE in the class of model-unbiased predictors. As the property of being unbiased under the model distribution is valid for non-linear indicators, the introduced BPs of unemployment rates are unbiased with lower MSE than other unbiased predictors. These are two very desirable optimality properties. However, we cannot calculate BPs because the model parameters are unknown, as it is the case for all model-based predictors. This is why, we substitute the model parameters by consistent estimators and calculate the EBP.

The EBP presents two drawback with respect to the BPs: (1) the model parameters and their estimates are not necessarily equal, and (2) the calculation of integrals requires approximations. Concerning the first drawback, we calculate REML

estimators of model parameters. Under regularity conditions, the REML estimators are consistent and therefore they asymptotically inherit the properties of the BPs. Concerning the second drawback, we consider different approximation methods to integrals. We further empirically investigate (1) and (2) by means of Simulations 1 and 2. From Simulation 1, we recommend using the GHQ method for approximating EBP of unemployment rates under BFH models. Simulation 2 considered several scenarios and showed that EBPs are superior to plug-in predictors for almost all considered settings, in particular with respect to bias-related measures.

The estimation of the MSE of the EBPs is challenging. We introduce a fully parametric bootstrap method and second method that decomposes the MSE in two summands of which only one is estimated by bootstrap. Simulation 3 investigates the behaviour of the two methods under different scenarios and recommends the second one.

The paper ends with an illustrative application to SLFS data, where the target is the estimation of unemployment rates by Spanish provinces, sex and age groups in the first trimester of 2021. We pay more attention to the unemployment of young persons, as it is of special political interest in Spain.

The micro data from the SLFS was downloaded from the INE web site. Unfortunately, this site does not provide the employed aggregated data at the level of the considered domains (province crossed by sex and age groups) from another data source. This is why we estimated the auxiliary variables from four waves of the last year, but treated them as exact values when applying the statistical methodology based on the fitted MFH model. The generalization of the MFH model to a measurement error MFH model for empirical best prediction of non-linear indicators is a subject for future research. Moreover, considering the paper's focus on predictive inference, the development of model selection criteria with emphasis on prediction accuracy rather than goodness of fit in model parameter estimation is another future research question.

Declaration of competing interest

All authors declare that they have no conflicts of interest.

CRediT authorship contribution statement

Jan Pablo Burgard: Writing – review & editing, Writing – original draft. **Joscha Krause:** Writing – review & editing, Writing – original draft. **Domingo Morales:** Writing – review & editing, Writing – original draft. **Anna-Lena Wölwer:** Writing – review & editing, Writing – original draft.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.ecosta.2024.09.001](https://doi.org/10.1016/j.ecosta.2024.09.001).

References

- Arima, S., Bell, W.R., Datta, G.S., Franco, C., Liseo, B., 2017. Multivariate Fay-Herriot Bayesian estimation of small area means under functional measurement error. *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 180 (4), 1191–1209. doi:[10.1111/rssa.12321](https://doi.org/10.1111/rssa.12321).
- Benavent, R., Morales, D., 2016. Multivariate Fay-Herriot models for small area estimation. *Computational Statistics & Data Analysis* 94, 372–390. doi:[10.1016/j.csda.2015.07.013](https://doi.org/10.1016/j.csda.2015.07.013).
- Benavent, R., Morales, D., 2021. Small area estimation under a temporal bivariate area-level linear mixed model with independent time effects. *Statistical Methods & Applications* 30 (1), 195–222. doi:[10.1007/s10260-020-00521-x](https://doi.org/10.1007/s10260-020-00521-x).
- Boubeta, M., Lombardía, M.J., Morales, D., 2016. Empirical best prediction under area-level poisson mixed models. *TEST* 25 (3), 548–569.
- Boubeta, M., Lombardía, M.J., Morales, D., 2017. Poisson mixed models for studying the poverty in small areas. *Computational Statistics & Data Analysis* 107, 32–47.
- Brown, G., Chambers, R., Heady, P., Heasman, D., 2001. Evaluation of small area estimation methods – an application to unemployment estimates from the UK LFS. In: *Proceedings of Statistics Canada Symposium*.
- Burgard, J.P., Esteban, M.D., Morales, D., Pérez, A., 2020. A Fay-Herriot model when auxiliary variables are measured with error. *TEST* 29 (1), 166–195. doi:[10.1007/s11749-019-00649-3](https://doi.org/10.1007/s11749-019-00649-3).
- Burgard, J.P., Esteban, M.D., Morales, D., Pérez, A., 2021. Small area estimation under a measurement error bivariate Fay-Herriot model. *Statistical Methods and Applications* 30, 79–108. doi:[10.1007/s10260-020-00515-9](https://doi.org/10.1007/s10260-020-00515-9).
- Burgard, J.P., Krause, J., Morales, D., 2022. A measurement error Rao-Yu model for regional prevalence estimation over time using uncertain data obtained from dependent survey estimates. *TEST* 31 (1), 204–234. doi:[10.1007/s11749-021-00776-w](https://doi.org/10.1007/s11749-021-00776-w).
- Burgard, J.P., Morales, D., Wölwer, A.-L., 2022. Small area estimation of socioeconomic indicators for sampled and unsampled domains. *ASTA Advances in Statistical Analysis* 106, 287–314. doi:[10.1007/s10182-021-00426-4](https://doi.org/10.1007/s10182-021-00426-4).
- Chandra, H., Salvati, N., 2018. Small area estimation of proportions under a spatial dependent aggregated level random effects model. *Communications in Statistics - Theory and Methods* 47 (5), 1234–1255. doi:[10.1080/03610926.2017.1317806](https://doi.org/10.1080/03610926.2017.1317806).
- Chandra, H., Salvati, N., Chambers, R., 2017. Small area prediction of counts under a non-stationary spatial model. *Spatial Statistics* 20, 30–56. doi:[10.1016/j.spasta.2017.01.004](https://doi.org/10.1016/j.spasta.2017.01.004).
- Chandra, H., Salvati, N., Chambers, R., 2018. Small area estimation under a spatially non-linear model. *Computational Statistics & Data Analysis* 126, 19–38. doi:[10.1016/j.csda.2018.04.002](https://doi.org/10.1016/j.csda.2018.04.002).
- Datta, G.S., Day, B., Basawa, I., 1999. Empirical best linear unbiased and empirical Bayes prediction in multivariate small area estimation. *Journal of Statistical Planning and Inference* 75 (2), 269–279. doi:[10.1016/S0378-3758\(98\)00147-5](https://doi.org/10.1016/S0378-3758(98)00147-5).
- Datta, G.S., Fay, R.E., Ghosh, M., 1991. Hierarchical and empirical multivariate Bayes analysis in small area estimation. In: *Proceedings of the Annual Research Conference*, pp. 63–79. U.S. Bureau of the Census, Washington, DC.
- Erciulescu, A.L., Cruze, N.B., Nandram, B., 2018. Benchmarking a triplet of official estimates. *Environmental and Ecological Statistics* 25, 523–547. doi:[10.1007/s10651-018-0416-4](https://doi.org/10.1007/s10651-018-0416-4).
- Erciulescu, A.L., Opsomer, J.D., 2019. A model-based approach to predict employee compensation components. In: *Joint Statistical Meetings. Government Statistics Section*, pp. 1601–1623. Alexandria, VA: American Statistical Association.

- Esteban, M.D., Lombardía, M.J., López-Vizcaíno, E., Morales, D., Pérez, A., 2020. Small area estimation of proportions under area-level compositional mixed models. *TEST* 29 (3), 793–818. doi:[10.1007/s11749-019-00688-w](https://doi.org/10.1007/s11749-019-00688-w).
- Faltys, O., Hobza, T., Morales, D., 2022. Small area estimation under area-level generalized linear mixed models. *Communications in Statistics - Simulation and Computation* 51 (12), 7404–7426. doi:[10.1080/03610918.2020.1836216](https://doi.org/10.1080/03610918.2020.1836216).
- Fay, R.E., 1987. Application of multivariate regression of small domain estimation. In: Platek, R., Rao, J.N.K., Särndal, C.E., Singh, M.P. (Eds.), *Small Area Statistics*. Wiley, New York, pp. 91–102.
- Fishman, G.S., 1996. Monte Carlo. Springer, New York doi:[10.1007/978-1-4757-2553-7](https://doi.org/10.1007/978-1-4757-2553-7).
- González-Manteiga, W., Lombardía, M.J., Molina, I., Morales, D., Santamaría, L., 2010. Small area estimation under Fay-Herriot models with non-parametric estimation of heteroscedasticity. *Statistical Modelling* 10 (2), 215–239. doi:[10.1177/1471082X0801000206](https://doi.org/10.1177/1471082X0801000206).
- González-Manteiga, W., Lombardía, M.J., Molina, I., Morales, D., Santamaría, L., 2008. Analytic and bootstrap approximations of prediction errors under a multivariate Fay-Herriot model. *Computational Statistics & Data Analysis* 52 (12), 5242–5252. doi:[10.1016/j.csda.2008.04.031](https://doi.org/10.1016/j.csda.2008.04.031).
- Hájek, J., 1964. Asymptotic theory of rejective sampling with varying probabilities from a finite population. *The Annals of Mathematical Statistics* 35 (4), 1491–1523.
- Jiang, J., 2003. Empirical best prediction for small-area inference based on generalized linear mixed models. *Journal of Statistical Planning and Inference* 111 (1–2), 117–127. doi:[10.1016/S0378-3758\(02\)00293-8](https://doi.org/10.1016/S0378-3758(02)00293-8).
- Jiang, J., Lahiri, P., 2001. Empirical best prediction for small area inference with binary data. *Annals of the Institute of Statistical Mathematics* 53 (2), 217–243.
- Jiang, J., Rao, J.S., Gu, Z., Nguyen, T., 2008. Fence methods for mixed model selection. *The Annals of Statistics* 36 (4), 1669–1692. doi:[10.1214/07-AOS517](https://doi.org/10.1214/07-AOS517).
- Krause, J., Burgard, J.P., Morales, D., 2022. Robust prediction of domain compositions from uncertain data using isometric logratio transformations in a penalized multivariate Fay-Herriot model. *Statistica Neerlandica* 76, 65–96. doi:[10.1111/stan.12253](https://doi.org/10.1111/stan.12253).
- Lange, K., 2010. *Numerical Analysis for Statisticians*, second ed. Springer, New York doi:[10.1007/978-1-4419-5945-4](https://doi.org/10.1007/978-1-4419-5945-4).
- López-Vizcaíno, E., Lombardía, M.J., Morales, D., 2013. Multinomial-based small area estimation of labour force indicators. *Statistical Modelling* 13 (2), 153–178.
- López-Vizcaíno, E., Lombardía, M.J., Morales, D., 2015. Small area estimation of labour force indicators under a multinomial model with correlated time and area effects. *Journal of the Royal Statistical Society. Series A (Statistics in Society)* 178 (3), 535–565.
- Marchetti, S., Secondi, L., 2017. Estimates of household consumption expenditure at provincial level in Italy by using small area estimation methods: “real” comparisons using purchasing power parities. *Social Indicators Research* 131 (1), 215–234. doi:[10.1007/s11205-016-1230-8](https://doi.org/10.1007/s11205-016-1230-8).
- Molina, I., Saei, A., José Lombardía, M., 2007. Small area estimates of labour force participation under a multinomial logit mixed model. *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 170 (4), 975–1000.
- Morales, D., Esteban, M.D., Pérez, A., Hobza, T., 2021. *A Course on Small Area Estimation and Mixed Models*. Springer doi:[10.1007/978-3-030-63757-6](https://doi.org/10.1007/978-3-030-63757-6).
- Morales, D., Pagliarella, M.C., Salvatore, R., 2015. Small area estimation of poverty indicators under partitioned area-level time models. *SORT* 39 (1), 19–34.
- Porter, A.T., Wikle, C.K., Holan, S.H., 2015. Small area estimation via multivariate Fay-Herriot models with latent spatial dependence. *Australian & New Zealand Journal of Statistics* 57 (1), 15–29.
- Press, W.H., Teukolsky, S.A., Wetterling, W.T., Flannery, B.P., 2007. *Numerical Recipes: The Art of Scientific Computing*, third ed. Cambridge University Press.
- Rao, J.N.K., 2014. Inferential issues in model-based small area estimation: Some new developments. *Statistics in Transition - new series* 16, 491–510.
- Rao, J.N.K., Molina, I., 2015. *Small area estimation*, second ed. John Wiley & Sons, Hoboken, NJ.
- Shang, J., Cavanaugh, J.E., Wright, F.T., 2008. A bayesian multiple comparison procedure for order-restricted mixed models. *International Statistical Review* 76 (2), 268–284.
- Smith, A.F.M., Gelfand, A.E., 1992. Bayesian statistics without tears: a sampling-resampling perspective. *The American Statistician* 46 (2), 84–88. doi:[10.1080/00031305.1992.10475856](https://doi.org/10.1080/00031305.1992.10475856).
- Stroud, A.H., Secrest, D., 1966. *Gaussian Quadrature Formulas*. Prentice-Hall, Englewood Cliffs, NJ.
- Sugasawa, S., Kubokawa, T., Rao, J., 2017. Small area estimation via unmatched sampling and linking models. *TEST* 27, 407–427.
- Sun, H., Berg, E., Zhu, Z., 2022. Bivariate small-area estimation for binary and Gaussian variables based on conditionally specified models. *Biometrics* 78, 1555–1565.
- Ubaidillah, A., Notodiputro, K.A., Kurnia, A., Mangku, I.W., 2019. Multivariate Fay-Herriot models for small area estimation with application to household consumption per capita expenditure in Indonesia. *Journal of Applied Statistics* 46 (15), 2845–2861. doi:[10.1080/02664763.2019.1615420](https://doi.org/10.1080/02664763.2019.1615420).
- Wood, J., 2008. On the covariance between related Horvitz-Thompson estimators. *Journal of Official Statistics* 24 (1), 53–78.
- Ybarra, L.M.R., Lohr, S.L., 2008. Small area estimation when auxiliary information is measured with error. *Biometrika* 95 (4), 919–931. doi:[10.1093/biomet/asn048](https://doi.org/10.1093/biomet/asn048).